

DHANUSH KUMAR

Full-Stack Developer · Backend & AI Engineer

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PROFESSIONAL SUMMARY

Final-year **Computer Science and Engineering** student (2026) with internship and production experience building AI-powered web systems, Agentic RAG pipelines, and high-throughput REST backends using FastAPI, Python, React, and Next.js. Skilled in LLM integration (LLaMA 3 via Ollama and Gemini), vector search (ChromaDB and pgvector), Azure cloud deployment, and full-stack application development from schema design to production.

TECHNICAL SKILLS

AI / ML:	Agentic RAG, LangGraph, LangChain, LLM Integration (LLaMA 3, Ollama, Gemini) Vector Search (ChromaDB, pgvector), Embeddings
Backend:	FastAPI, REST APIs, PostgreSQL, SQLAlchemy, JWT/RBAC, WebAuthn/FIDO2, Async Python, Redis, ARQ
Frontend:	React.js, Next.js 14, Tailwind CSS, shadcn/ui, Responsive UI, HTML, CSS, JavaScript
DevOps/Tools:	Git/GitHub, Docker, Azure (Cloud Deployment, Azure DevOps Pipelines), Linux, Nginx, GitHub Actions, CI/CD
Languages:	Python, JavaScript (ES2023+), Java Design: Figma, Wireframing, Prototyping

INTERNSHIP EXPERIENCE

Web Development Intern · TIBOS Solutions and Services Pvt Ltd

Feb 2026 – May 2026

- Contributed to a production CRM platform; collaborated on debugging, testing, and deployment to Azure Cloud through Azure DevOps CI/CD pipelines.
- Gained hands-on experience with real-time client requirements, sprint workflows, and enterprise-grade software delivery.

Web Development Intern · SkillUpU

Sep 2024 – Oct 2024

- Developed and maintained responsive web pages using HTML, CSS, and JavaScript; improved UI/UX and site performance in a team environment.

Backend Development Intern · CodTech IT Solutions

Sep 2024 – Nov 2024

- Designed user-friendly web interfaces and developed real-time chat APIs using Socket.IO and Flask, enabling seamless bidirectional communication between users.

ENGINEERING PROJECTS

RAG-Powered AI Learning Platform | FastAPI · LLaMA3/Ollama · Gemini Embeddings · ChromaDB · pgvector · Python

- Architected an end-to-end Agentic RAG pipeline: document ingestion → chunking → embedding generation → ChromaDB vector indexing → semantic retrieval → LLM response, achieving sub-800ms answer latency on commodity hardware.
- Built async FastAPI backend with pgvector cosine-distance ranking, achieving 90%+ retrieval precision vs. keyword search; integrated locally hosted LLaMA3 via Ollama, eliminating external API dependency and cutting per-query cost to zero.

Inventory Billing & Digital Store System | React · FastAPI · PostgreSQL · Barcode/QR Engine · Tailwind CSS

- Deployed full-stack retail system with barcode/QR scanning, automated billing, and real-time inventory tracking — reducing checkout time by ~60%; built index-optimized FastAPI backend sustaining sub-100ms response under concurrent sessions.
- Delivered admin dashboard with sales analytics and CSV export, saving ~5 hours/week of manual reconciliation work.

CRM & Order Management System | React · Next.js · FastAPI · PostgreSQL · Redis · JWT/RBAC · ARQ

- Architected scalable enterprise CRM centralizing customer management, subscriptions, payments, and distributor operations using FastAPI, PostgreSQL, and Redis with ARQ background workers for real-time analytics.
- Implemented RBAC, JWT auth, bulk Excel import/export, and automated payment tracking — significantly reducing administrative overhead.

EDUCATION

B.E. in Computer Science and Engineering – Latha Mathavan Engineering College

Expected June 2026

Coursework: Data Structures & Algorithms, Database Systems, Artificial Intelligence, Web Development, Computer Networks

AWARDS & ACHIEVEMENTS

- Cyberthan Hackathon 2025 – 1st Place:** Built WebAuthn/FIDO2 passwordless auth system; top finish in cybersecurity category at Kumaraguru College of Engineering.
- Codify Challenge:** Ranked among top performers in competitive algorithmic coding contest at Kamaraj Engineering College.
- Technical Speaker – PSR Engineering College:** Delivered a session on Cryptocurrency and Blockchain to an audience of 100+ students.